higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## CARPENTRY AND ROOFWORK
## NQF LEVEL 3

(12030333)

**21 November 2019 (X-Paper)**
**09:00–12:00**

**This question paper consists of 6 pages.**

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**TIME: 3 HOURS**
**MARKS: 100**

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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## QUESTION 1: HIPPED ROOFS

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write 'True' or 'False' next to the question number (1.1.1–1.1.8) in the ANSWER BOOK.

1.1.1 The ridge on a rectangular hipped roof is parallel to the eaves.

1.1.2 A roof with a span of more than 6 metres must be designed by a specialist.

1.1.3 A discrepancy of 20 millimetres is acceptable in the external dimensions of a building.

1.1.4 Roof trusses should always be stacked against the exterior walls before being raised.

1.1.5 A gable roof and a hipped roof with the same pitch experience the same wind force.

1.1.6 Jack rafters are supporting members found in all roof structures.

1.1.7 A deemed-to-satisfy roof is when the manufacturing process is based on the carpenter's experience and knowledge of SANS 10400 and SANS 10243.

1.1.8 When inspecting a roof, an inspector does not check all specifications in the layout if they have been complied with.

(8 × 1) (8)

1.2 A hipped roof must be rationally designed.

Which THREE key factors are taken into account during the design process? (3)

1.3 Give TWO examples of how poor workmanship can affect the stability of a roof. (2)

1.4 Give ONE reason why a girder truss is assembled on the ground before being raised. (2)

1.5 Bolts, washers and nails are commonly used when assembling a girder truss.

What are the dimensions of the bolts and nails respectively? (2)

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1.6 The truss manufacturer labels the trusses and roof components to simplify their identification on site.

What do the following labels indicate?

1.6.1 HG

1.6.2 GT

1.6.3 BC

(3 × 1) (3)

1.7 Explain how the absence of soffits at the eaves affects the fixing of plastic underlay. (4)

1.8 Give FOUR reasons why, compared to a gabled roof, it is expensive to erect a hipped roof. (4)

1.9 Name the TWO points in the roof framework to which the jack rafters are connected. (2)

**[30]**

## QUESTION 2: ERECT AN L-SHAPED ROOF

2.1 Complete the following sentences by using the words in the list below. Write only the words next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

| |
|---|
| safety harness; diagonal bracing; wall plate; tilting batten; double wedges |

2.1.1 Strips of hardboard or … can be placed under the tie beam to level the trusses.

2.1.2 A … is a timber member laid flat on the supporting wall usually flush with the inner skin of the wall.

2.1.3 The … is thicker because it compensates for the first row of tiles not having a preceding row of battens under it.

2.1.4 It is compulsory for roof erectors to wear a … as the risk of falling is always present.

2.1.5 The hips and jack rafters at either end of the roof stabilise the roof and eliminate the need for …

(5 × 1) (5)

2.2 Draw the plan of an L-shaped roof. One end of the roof must be hipped and the other end gabled.

The following details must be clearly shown and labelled:

2.2.1 Gable end

2.2.2 Hipped end

2.2.3 Hip girder

2.2.4 Jack truss

2.2.5 Valley

2.2.6 Ridge

2.2.7 Full truss

(2 × 7) (14)

2.3 Give THREE reasons why the valley of a roof is vulnerable to damage. (3)

2.4 List the FOUR steps to be taken when calculating the truss spacing on an L-shaped mono-pitch roof. (2 × 4) (8)

**[30]**

## QUESTION 3: INSPECTION OF ROOF STRUCTURES

3.1 Explain the roles and responsibilities of the following personnel during a roof inspection.

3.1.1 Building owner

3.1.2 Roof erector

3.1.3 Occupant of the house

3.1.4 Designing engineer

(2 × 4) (8)

3.2 List FOUR conditions that should be satisfied during a roof inspection. (4)

3.3 Certain allowances are made when inspecting a roof. With this in mind, fill in the missing dimensions in the following statements :

3.3.1 The bow between purlins or battens shall not be more than … millimetres.

3.3.2 NO section of the roof shall be more than … millimetres out of plumb.

3.3.3 Any member more than … millimetres skew should be reported.

(3 × 1) (3)

**[15]**

## QUESTION 4: CLAD A TIMBER ROOF STRUCTURE

4.1 Make a neat line diagram to show the overlap between two IBR roofing sheets. Use an arrow to indicate the direction of the prevailing (rain) wind. (6)

4. 2 Name the type of roofing material used on:

4.2.1 Class 'A' roof

4.2.2 Class 'B' roof

4.2.3 Class 'C' roof

(3 × 1) (3)

4.3 List FOUR ways in which the choice of the roof covering material can affect the design of the roof. (4)

4.4 Give TWO reasons why it is compulsory to install an under-tile membrane (underlay) on tiled roofs with pitches between 17,5° and 26°. (2)

4.5 Give FIVE factors that could affect the fixing of roof tiles. (5)

4.6 Bituminous aluminium paint is applied to roofing sheets to prevent white rust.

Name TWO other reasons why this product can also be extremely useful.

(2 × 2) (4)

4.7 Where is mortar used when laying tiles on a roof? (1)

**[25]**

**TOTAL: 100**

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